# Math Kangaroo Practice

Grade 3-4 · 20 questions · harder · seed 14083

***Part A: Each question is worth 3 points***

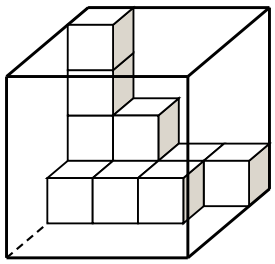
**1.** A street has 132 houses, numbered from 1 to 132. Tom writes all the house numbers on cards, one digit on each card.
How many cards does Tom need?

***A.*** 396 ***B.*** 297 ***C.*** 288 ***D.*** 279 ***E.*** 255

**2.** Alex has the cards 1, 7 and 8. Alex must use every card exactly once, with +, –, ×, ÷ and brackets. Which of these numbers can Alex get?

***A.*** 52 ***B.*** 51 ***C.*** 48 ***D.*** 41 ***E.*** 40

**3.** Mia has a glass aquarium shaped like a cube, 4 dm long, 4 dm wide and 4 dm tall. Mia has started to put cubes with edges of 1 dm into it, as in the picture. Every cube stands on the bottom or on another cube.

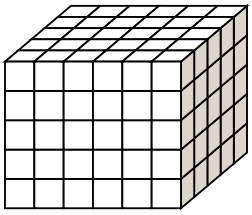


At most how many more such cubes can Mia put into the aquarium?

***A.*** 11 ***B.*** 49 ***C.*** 52 ***D.*** 53 ***E.*** 55

***Part B: Each question is worth 4 points***

**4.** A brick 6 cubes long, 5 cubes deep and 5 cubes tall is made of small cubes. It is painted on the outside, also underneath, and then taken apart.



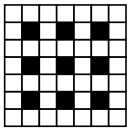
How many small cubes have exactly 1 painted face?

***A.*** 66 ***B.*** 69 ***C.*** 70 ***D.*** 75 ***E.*** 79

**5.** The diagrams below are the 1st, 2nd and 3rd of a pattern. How many cells in all will the 8th diagram have?

8 white cells

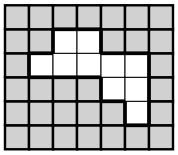
21 white cells

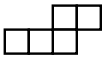


40 white cells

***A.*** 324 ***B.*** 305 ***C.*** 289 ***D.*** 256 ***E.*** 64

**6.** Two pieces that are exactly the same cover the empty area. Which piece is it? The pieces may be turned and flipped over.



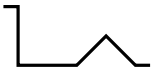
***A.*** ***B.*** ***C.*** ***D.***  ***E.***

**7.** Finn's snake toy is 22 dm 9 cm long. Dan's snake toy is 1850 mm long.
How many centimetres longer is Finn's snake?

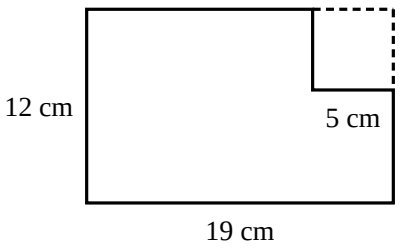
***A.*** 440 cm ***B.*** 53 cm ***C.*** 44 cm ***D.*** 42 cm ***E.*** 34 cm

**8.** This is the outline of a town seen from the river at sunset.

Which piece **can** be part of the outline?

***A.***  ***B.*** ***C.*** ***D.*** ***E.***

**9.** A rectangle is 19 cm long and 12 cm wide. A square with side 5 cm is cut off one corner.




What is the perimeter of the remaining shape?

***A.*** 42 cm ***B.*** 52 cm ***C.*** 62 cm ***D.*** 72 cm ***E.*** 82 cm

**10.** Omar was born on February 13, 2012 and Leo on December 28, 2013.
How many years old is Leo on November 15, 2019?

***A.*** 9 ***B.*** 8 ***C.*** 7 ***D.*** 6 ***E.*** 5

**11.** Tom thinks of a number, divides by 5, then multiplies by 2, then takes away 1, and gets 7.
Which number did Tom think of?

***A.*** 22 ***B.*** 20 ***C.*** 19 ***D.*** 15 ***E.*** 7

**12.** Only the even numbers from 2 to 120 are written down: 2, 4, 6, ..., 120.
How many digits are written?

***A.*** 60 ***B.*** 127 ***C.*** 129 ***D.*** 143 ***E.*** 146

***Part C: Each question is worth 5 points***

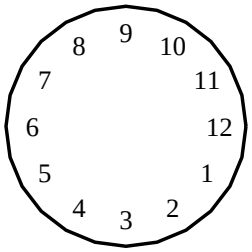
**13.** Five children, Clara, Grace, Emma, Ben and David, sit in a row of 5 chairs at the cinema. Clara and Grace are best friends and want to sit next to each other.
In how many different ways can the children sit?

***A.*** 120 ***B.*** 72 ***C.*** 48 ***D.*** 24 ***E.*** 4

**14.** A bar code is made of 22 black and white bars taking turns. It starts with a black bar and ends with a **white** bar. Each black bar is wide or narrow. There are 4 more white bars than wide black bars.
How many narrow black bars are there?

***A.*** 8 ***B.*** 7 ***C.*** 6 ***D.*** 5 ***E.*** 4

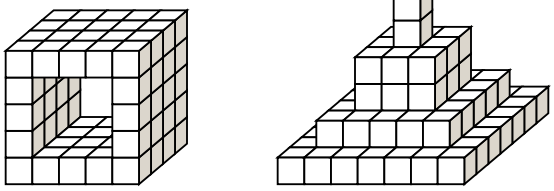
**15.** A round cake with the numbers 1 to 12 around its edge is broken by one straight crack into two pieces. The sums of the numbers on the two pieces are equal.




Where does the crack go?

***A.*** between 9 and 10, and between 3 and 4
***B.*** between 8 and 9, and between 3 and 4
***C.*** between 4 and 5, and between 7 and 8
***D.*** between 8 and 9, and between 2 and 3
***E.*** between 2 and 3, and between 7 and 8

**16.** Tom has the tunnel on the left. Tom takes it apart to build the pyramid on the right, but there are not enough cubes. Nothing is hollow inside the pyramid.



tunnel pyramid

How many more cubes does Tom need?

***A.*** 78 ***B.*** 46 ***C.*** 30 ***D.*** 20 ***E.*** 14

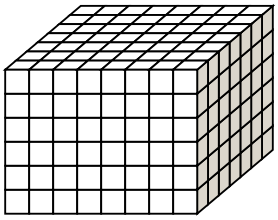
**17.** The odometer of a scooter shows 1469 km. Each digit is bigger than the digit before it. After how many more kilometres will this happen again?

***A.*** 10 ***B.*** 9 ***C.*** 8 ***D.*** 7 ***E.*** 1

**18.** How many three-digit numbers have digits that add up to 6?

***A.*** 42 ***B.*** 26 ***C.*** 23 ***D.*** 22 ***E.*** 21

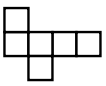
**19.** Chris builds a brick 8 cubes long, 7 cubes deep and 6 cubes tall from red and blue cubes. All the cubes you can see from the outside are red. The cubes just below them are **also** red, so on every side (top, bottom, front, back, left and right) there are two layers of red cubes. All the other cubes, in the middle, are blue.



How many blue cubes are there?

***A.*** 12 ***B.*** 24 ***C.*** 60 ***D.*** 120 ***E.*** 312

**20.** Mia has two copies of the same piece. With which piece can she make a rectangle? Pieces may be turned and flipped over.

***A.*** ***B.*** ***C.***  ***D.*** ***E.***

# Answer Key

| | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1. | **C** | **2.** | **C** | **3.** | **D** | **4.** | **A** | **5.** | **C** | **6.** | **E** |
| 7. | **C** | **8.** | **B** | **9.** | **C** | **10.** | **E** | **11.** | **B** | **12.** | **B** |
| 13. | **C** | **14.** | **E** | **15.** | **A** | **16.** | **C** | **17.** | **B** | **18.** | **E** |
| 19. | **B** | **20.** | **E** | | | | | | | | |

### *Solutions*

**1. C** (288) — 9 × 1 + 90 × 2 + 33 × 3 = 288. [2003 Q16 · digits_for_pages]

**2. C** (48) — (7 – 1) × 8 = 48. None of the other numbers can be made. [2002 Q21 · which_can]

**3. D** (53) — The aquarium holds 4 × 4 × 4 = 64 cubes. There are already 11 inside (count column by column, also the hidden ones). 64 – 11 = 53. [2007 Q14 · how_many_more]

**4. A** (66) — The middle part of each face: 2 × (4 × 3 + 4 × 3 + 3 × 3) = 66. [2007 Q17 · painted_brick]

**5. C** (289) — The 8th diagram is 17 × 17 = 289 cells, with 8 × 8 = 64 black ones. [2007 Q21 · black_and_total]

**6. E** — The empty area has 10 squares, so each piece has 5; every option has 5. Only two copies of piece E fit together into the shape of the hole. [2003 Q18 · two_copies]

**7. C** (44 cm) — 22 dm 9 cm = 229 cm and 1850 mm = 185 cm. 229 – 185 = 44 cm. [2003 Q10 · how_much_longer]

**8. B** — Only this piece is found in the line. The others don't fit — A: its flat part between the two features has the wrong length; C: it has a pointed roof where the line has a round dome; D: it has two small bumps where the line has a pointed roof; E: it has the right shapes but in the wrong order. [2002 Q7 · can_belong]

**9. C** (62 cm) — The two new edges (5 cm and 5 cm) replace the two removed pieces of the outer edge, which were also 5 cm and 5 cm. So the perimeter is the same as the rectangle's: 2 × (19 + 12) = 62 cm. [2002 Q6 · corner_cut]

**10. E** (5) — 2019 – 2013 = 6, but Leo's birthday (December 28) has not come yet that year, so Leo is 5. [2007 Q12 · age_on_day]

**11. B** (20) — Go backwards: 7 + 1 = 8, 8 ÷ 2 = 4, 4 × 5 = 20. [2007 Q5 · think_of_a_number]

**12. B** (127) — Even one-digit numbers: 4; even two-digit numbers: 45; even three-digit numbers: 11. 4 × 1 + 45 × 2 + 11 × 3 = 127. [2007 Q23 · even_only]

**13. C** (48) — Think of Clara and Grace as one block: the block and the other 3 make 4 things to order, 4 × 3 × 2 × 1 = 24 ways. Inside the block Clara and Grace can swap, so 2 × 24 = 48. [2002 Q22 · side_by_side]

**14. E** (4) — Starting black and ending white, the bars come in pairs: 11 black and 11 white. Wide: 11 – 4 = 7. Narrow: 11 – 7 = 4. [2003 Q17 · ends_differ]

**15. A** (between 9 and 10, and between 3 and 4) — Total 78, so each piece has 39: 10+11+12+1+2+3 = 39 and 4+5+6+7+8+9 = 39. [2002 Q11 · straight_crack]

**16. C** (30) — Tunnel: 16 × 4 = 64 cubes. Pyramid: 7 × 7 + 5 × 5 + 3 × 3 × 2 + 1 × 1 × 2 = 94. Missing: 94 – 64 = 30. [2002 Q20 · not_enough]

**17. B** (9) — Keep as many digits on the left as possible. The next such number is 1478: 1478 – 1469 = 9. [2002 Q24 · going_up]

**18. E** (21) — Count by the first digit: 6 + 5 + 4 + 3 + 2 + 1 = 21. [2007 Q4 · count_with_sum]

**19. B** (24) — Each length loses two red cubes at each end, so it drops by 4: 4 × 3 × 2 = 24. [2003 Q9 · thick_wall]

**20. E** — Two copies have 12 squares, so the rectangle is 1 × 12 or 2 × 6 or 3 × 4. Only two copies of piece E fit together like that. [2007 Q8 · two_copies]